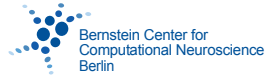


The Decoding Toolbox (TDT): An easy-to-use decoding package for fMRI data



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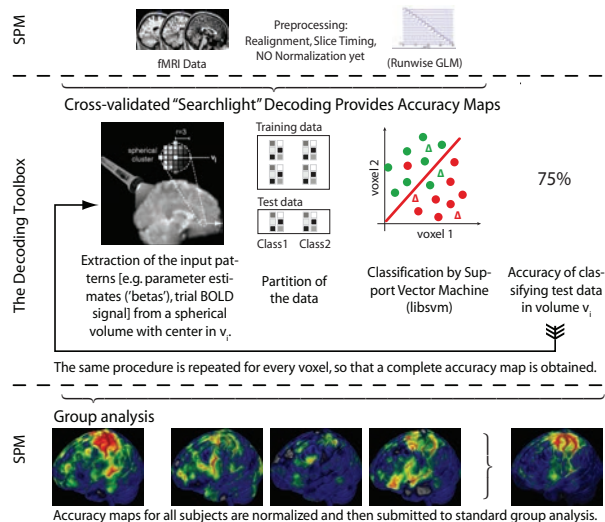
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Introduction

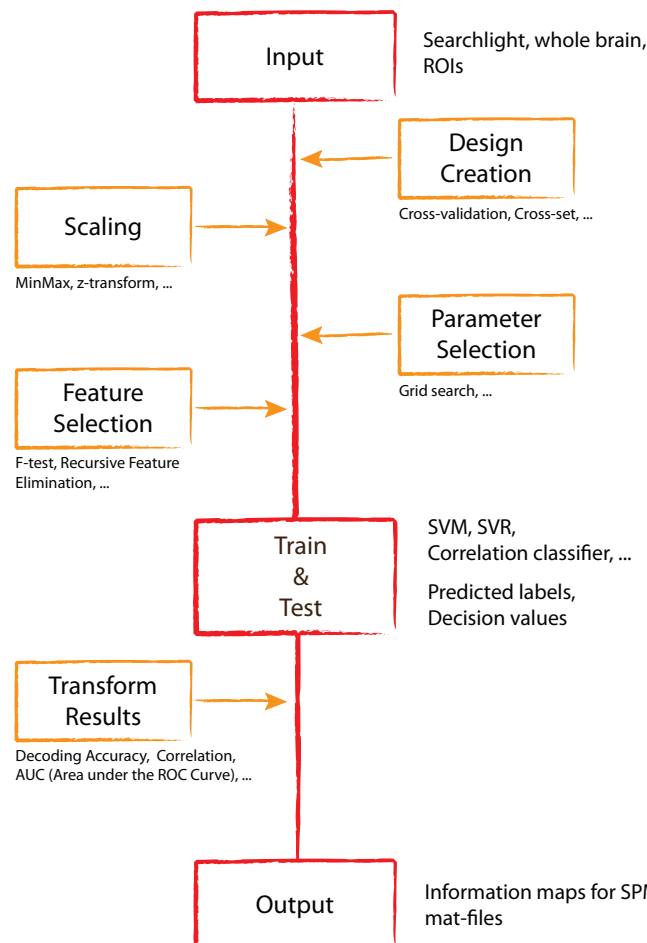
'Decoding' of brain data (e.g. fMRI) has recently received tremendous interest, among others due to its increased sensitivity in detecting meaningful patterns of brain activity. However, few standard software packages for decoding exist, and thus nearly all existing publications make use of different – often unpublished – analysis pipelines. Drawbacks of this include the difficulty to reproduce results, possibly unnoticed programming errors, or the problem for new users to quickly take advantage of decoding techniques for their analyses.

Here, we present a new decoding toolbox that is flexible, easy to use and yet powerful. The toolbox is compatible with – but not limited to – SPM, the most common fMRI data analysis software. The toolbox can be used by any researcher with basic knowledge in fMRI data analysis and minimal programming skills, but also allows maximal flexibility in the application of complex decoding designs for more experienced researchers.

A Standard Decoding Analysis

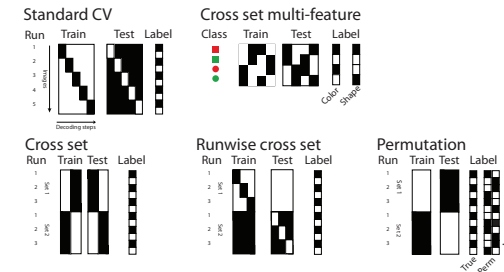


Toolbox Workflow



Example Decoding Designs

Many designs can be created automatically



Killer Features

- 1. SIMPLICITY:** The Decoding Toolbox is very easy to use. Use your SPM.mat to create a full leave-one-run out cross-validation in one line of code. Try it out!
- 2. LOGGING, WARNING & ERRORS:** We cared a lot about the output, so that you know what went on when something went wrong. Different levels of outputs allow you to get just the amount of information that you need, from simply "do it" to the smallest detail.
- 3. GENERALITY:** Do searchlight decoding, whole brain decoding or ROI without changing the rest of your code!
- 4. FLEXIBILITY:** The modular structure allows easily setup all sorts of classification designs. Or simply extend it by using plugging in your own algorithms and measures.
- 5. READABILITY:** During development, we cared a lot about code quality. We believe that code is written for humans, not for machines. This should make it easy for you to read and understand what's going on or to adjust the code to your needs.
- 6. SPEED:** Be fast! You are not only fast writing the code: The toolbox is also fast running it. Optimized data handling & functions speed up processing a lot compared to using standard matlab-functions and self-written scripts.

Contact & How to get TDT

If you like the toolbox, please contact us:
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